

PROJECT TRICORDER

MK I · FIELD OPERATIONS MANUAL · REV 0.6.14

On-device multimodal field instrument. Inquiry, vision, image synthesis, neural speech in/out, offline translator, audio scene analysis with soundscape watch, avian and flora/fauna classification, sentry weapon/violence watch, IPL abort, artificial horizon with G-meter and GPS speed, Celestrak/SGP4 satellite plot, major-star map, SOL space weather and inner-system orbits, NWS earth forecast, NOAA / flight / civic watch, append-only field log, optional tricorder-to-tricorder mesh.

Jim Ames, Claude and Grok · N2NHU Labs · Newburgh, New York 12550. Implemented as RENT-A-HAL Android Edition, Project Tricorder Mk I. Software 0.6.14. This revision supersedes 0.6.8 (6 September 2026).

1. Purpose and scope

This is the operator and installer document for Project Tricorder Mk I through software 0.6.14. It covers face use, what changed after Rev 0.6.8, theory of the new watches, limits, and troubleshooting. Features documented in 0.6.4–0.6.8 remain current unless a later section says otherwise.

NAVIGATION WARNING. The artificial horizon, G-meter, GPS speed, satellite plot, star map, SOL orbits, and HF/VHF/UHF forecasts are NOT certified by FAA, NASA, EASA, ICAO, or any civil aviation or space authority. They MUST NEVER be used as primary or backup instruments to fly, land, or navigate any aircraft, spacecraft, vessel, or vehicle.

SENTRY / AUDIO WATCH WARNING. Weapon, gunfire, and conflict alerts are keyword and classifier hits on SmolVLM captions or YAMNet AudioSet labels. They are not a weapons detector, not a 911 substitute, and not a legal identification of a firearm. Covered carry, a dark frame, fireworks, and talk radio can all miss or false.

2. What changed since Rev 0.6.8

Rev 0.6.8 (6 Sep 2026) closed the SOL SYSTEM face. Software 0.6.9–0.6.14 added the field-sentry and audio-sentry layer:

Build	Change
0.6.9	Sentry keyword watch on vision captions: sure weapon / possible weapon / violence / animal / rain. Robot voice + klaxon or chime. Not Kokoro.
0.6.10	Sentry III defaults to Continuous. Vision captions every dwell; motion no longer gates the look.
0.6.11	Watch also fires on Sensor II if the caption names a gun or possible weapon. Motion metric is changed-pixel fraction (~7%).
0.6.12	Motion cannot overwrite a weapon caption or flush the robot voice. Motion announce only if no keyword hit, 20 s cool.
0.6.13	IPL lamp + abort slider. Separate Kokoro-kill slider.

	Kokoro splits text over 80 characters. Settings dwell sliders. First camera frame sent as soon as the sensor is ready.
0.6.14	Audio IV soundscape watch: gunfire, scream/shout, siren, speech, entertainment, road vehicles, air/rail/ship, named animals. Robot voice. Logged.

3. Face controls (0.6.14)

Control	Function
I Inquiry	Gemma chat. Imagine keywords. COMPUTER easter egg. Exact HELP opens the spoken briefing and Help sheet.
II Sensor	Camera + SmolVLM + flora/fauna. Weapon/possible-weapon watch on the caption. First frame as soon as the camera is ready; dwell is the gap between looks.
III Sentry	Continuous by default. Captions every dwell. Motion is logged/spoken only if the caption has no watch hit. Sure weapon: klaxon + WEAPON DETECTED ×3. Possible weapon/violence: chime + short phrase. Animal/rain: robot line. Hold after a look unless Continuous stays on.
IV Audio	YAMNet duty cycle, BirdNET when the clip looks avian, context watches if checked. Soundscape watch (see §6). Robot voice, not Kokoro.
T Translator	Offline speech-to-speech. SOURCE / TARGET wheels. AUTO swap after each utterance unless toggled red.
H Horizon	Green = tape on (idle + Audio). Red = off so the CRT can show the log. G-meter and GPS speed on the tape. Not flight-rated.
S Sat/Stars	Cycles satellites → stars → off. Polar live plot from Celestrak + SGP4.
O Sol	Cycles space weather + schematic orbits → earth forecast → off. See Rev 0.6.8 §5–§7.
Idle / Active / Alert / IPL	Status bank. IPL green while an abort is running, red after it trips.
IPL slider	Throw to 1: abort in-flight chat, vision, Imagine, voice, and translate HTTP. Lamp green then latches red. Slider springs back.
Kokoro slider	Throw to 1: kill only Kokoro playback and queued chunks. Does not abort vision or Imagine.
Chat / Vision / Image / Voice / Translate lamps	Green idle, amber thinking, red unreachable.
L / E / V / P	Log review slider, officer audio note, video log, play current entry.
Hold / Continuous	Hold locks the last frame. Continuous keeps looking

	(default on III). Continuous machine vision (no Kokoro) writes captions without flooding TTS.
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4. Settings added after 0.6.8

Settings → Sensor / Sentry dwell. Two sliders. Sensor default 3 s (range 2–15). Sentry default 3 s (range 2–12). These are the seconds between looks after the first send. The first JPEG leaves as soon as the camera reports readyState ≥ 2. A long first look (often 8–15 s on Snapdragon 888) is SmolVLM inference, not a hidden pre-roll.

Image quality/speed sliders from earlier revisions are unchanged: step count and size (256 / 384 / 512). Turbo checkpoint, cfg-scale 1.

5. Sentry watch — theory of operations

5.1 Why it exists

III on a rock at a cave mouth is the intended use. The operator is not holding the phone. Continuous vision writes a caption every dwell. A second pass reads that caption for a short keyword list. Alerts use Android TextToSpeech (robot voice) and ToneGenerator. Kokoro is reserved for inquiry and long prose so a klaxon is not queued behind a 40-second speech job.

5.2 Tiers

Tier	Example tokens	Voice	Tone	Log kind
Sure weapon	pistol, handgun, revolver, rifle, shotgun, firearm, Glock, AK-47, AR-15, knife, dagger, machete, sword	WEAPON DETECTED ×3	Klaxon	sentry_weapon
Possible weapon	bat, pipe, chain, club, mace (dual-use)	possible weapon detected	Seat-belt chime	sentry_possible
Possible violence	fighting, striking, hitting, kicking, violence	possible violence detected	Chime	sentry_possible
Scene	animal, rain (and similar caption words)	animal detected / rain detected	None	sentry_scene
Motion only	pixel-change ≥ ~7% and no keyword	motion detected (20 s cool)	None in continuous	sensor / sentry

5.3 Motion is not a gate

Builds before 0.6.10 waited for an 18% mean-gray change before asking vision. That hid still objects (a pistol on a table). From 0.6.10, III captions every loop. Motion is a side metric: fraction of 64×48 pixels whose gray value moved more than 22 counts. From 0.6.12, a motion line cannot replace a weapon caption on the CRT or flush the robot utterance.

5.4 Sensor II

Pointing II at an object also runs the sure/possible weapon watch. Motion and rain lines stay III-only so a living-room scan does not announce “motion detected” every dwell.

5.5 Limits

The model must print a token the regex knows. “Metal object in a holster,” a covered firearm, or a dark cave will stay quiet. A toy caption that contains “gun” can trip the sure tier unless it matches the water/toy/nail/airsoft exception. Treat the klaxon as “the caption named a weapon,” not “a weapon is present.”

6. Audio soundscape watch (0.6.14)

IV already classified the clip with YAMNet (521 AudioSet classes) and optionally BirdNET. 0.6.14 maps the top sixteen labels onto a priority list and speaks the highest hit with the same robot voice as sentry. Kokoro is not used here. One spoken line per listen window.

Priority	YAMNet-ish labels	Spoken line	Tone	Cool	Log kind
1	Gunshot, gunfire, machine gun, artillery, fusillade, cap gun	possible gunfire audio detected ×3	Klaxon	20 s	audio_gunfire
2	Screaming, shout, yell, wail, battle cry	possible human conflict emergency detected: screaming, shouting	Chime	18 s	audio_conflict
3	Siren, ambulance, police car, fire engine, emergency vehicle	emergency vehicles active in the area ×3	Chime	20 s	audio_siren
4	Aircraft, airplane, helicopter, jet engine	air traffic detected	—	25 s	audio_air

5	Train, railroad, subway, train horn	rail traffic detected	—	25 s	audio_rail
6	Ship, boat, motorboat, foghorn	ship traffic detected	—	25 s	audio_ship
7	Car, truck, bus, motorcycle, traffic / roadway noise	vehicles detected	—	25 s	audio_vehicle
8	Dog/bark, cat/meow, bird, cow, horse, frog, insect...	dog detected (or cat, bird, ...)	—	20 s	audio_animal
9	Speech, conversation, narration	humans detected ×3	—	50 s	audio_speech
10	Music, song, radio, television	entertainment device detected	—	40 s	audio_media

Speech and entertainment use long cooldowns so a campsite radio does not talk over itself. BirdNET still runs when YAMNet looks avian; species lines stay on the CRT and in the bird log. Fireworks can look like gunfire. Talk radio can look like humans. A distant jet can look like air traffic. Log review (L) is the record, not the voice.

6.1 Duty cycle

Audio sliders still set listen dwell and sleep interval. The mic sleeps between windows. holdForVoice pauses the classifier while Kokoro or robot TTS is speaking so the unit does not classify its own voice.

7. IPL and Kokoro kill (0.6.13)

IPL (Initial Program Load) sits in the Idle / Active / Alert bank. Throw the IPL slider: in-flight HTTP to chat (:8080), vision (:8081), Imagine (:8188), Kokoro (:5000), and the translator pipeline are aborted. The lamp is green during the abort and latches red when the reset has tripped. The slider returns to 0.

The Kokoro slider kills only playback: increments the generation counter, disconnects the current /v1/audio/speech POST, stops MediaPlayer, and drops queued 80-character chunks. Vision and Imagine keep running.

Kokoro text longer than 80 characters is split on sentence then word boundaries. The first chunk synthesizes immediately so a long briefing is not silent for a minute. Decimal points in USS, pressure, and similar strings are still stripped before TTS so “U.S.S.” does not stall the phonemizer.

8. Field log kinds added after 0.6.8

Append-only JSONL with UTC and last GPS when available. New kinds: sentry_weapon, sentry_possible, sentry_scene, audio_gunfire, audio_conflict, audio_siren, audio_air, audio_rail, audio_ship, audio_vehicle, audio_animal, audio_speech, audio_media. Existing kinds (sensor, sentry, bird, flora, audio, officer, video) are unchanged. L slides the timeline. P plays officer and video entries.

9. Architecture recap (unchanged core)

APK is a Kotlin WebView shell (TricorderBridge). Termux runs four local servers that must all stay up: llama-server Gemma chat :8080, SmolVLM vision :8081, sd-server Imagine :8188, Kokoro ONNX :5000.

~/bin/tricorder start launches them plus sshd on 8022. YAMNet, BirdNET, Coral flora/fauna, and ML Kit translation run in-process in the APK. Nearby Connections is the camp mesh. ContextWatch polls NOAA CAP, OpenSky, and civic/USGS feeds when IV is armed and Wi-Fi is up.

10. SOL / sky / horizon (from 0.6.4–0.6.8)

Unchanged. O cycles space (SWPC + HamQSL HF/VHF + derived UHF aurora + schematic Mercury–Saturn orbits) → scrollable NWS six-hour + three-day earth forecast → off. S cycles Celestrak groups → major stars → off. H toggles the uncertified horizon. Fetch SOL wx on the SOL bar if HF rows are dashes after an upgrade.

11. Install and first run

Sideload the APK. Grant camera, microphone, location, nearby devices as prompted. Install Termux (F-Droid). Place the compiled binaries and GGUF/ONNX weights as in the project zip. ~/bin/tricorder start, then tricorder status until chat, vision, image, and tts report up. On first launch with earth link (8.8.8.8), the APK fetches BirdNET, flora/fauna, and a translator pair if they are missing, and pulls TLE + SOL snapshots. If Termux is absent the CRT says so; I–Imagine and Kokoro will stay red.

12. Troubleshooting

Symptom	Likely cause	Action
Weapon caption, no klaxon (pre-0.6.11)	Watch only ran in continuous III	Install 0.6.12+; use II or III
Klaxon but no speech; CRT says motion detected	Motion flushed TTS (0.6.11)	Install 0.6.12+
First II/III look takes ~10 s	SmolVLM cold inference, not a timer	Expected on SM888; dwell only affects the next look
Kokoro silent on a long paragraph	Single 2 kB job or collision	0.6.13 chunks at 80 chars; throw Kokoro slider to abort a stuck job
All backend lamps red, Termux up	127.0.0.1 probes failing or services died	~/bin/tricorder status; start; do not stop 8188 to free a port
Audio armed, no analysis	Build after 0.3.2 without the 0.3.2 listen path (fixed later)	Use 0.6.14; confirm YAMNet model in APK assets

Audio calls everything speech	Operator voice or Kokoro leaking into the clip	holdForVoice should pause the mic; raise speech cooldown; point the mic away
Fireworks announced as gunfire	YAMNet AudioSet overlap	Expected limit; check the log score and label
IPL slider does nothing	Build older than 0.6.13	Install 0.6.13+
HF rows dashes / earth pane empty	Stale SOL snapshot or no GPS	Wi-Fi, location on, Fetch SOL wx
Horizon needs a 90° rotate	Addressed in later 0.6.x portrait remap; if still wrong, hold portrait	Portrait gravity remap is in 0.6.4+

13. Competing products (unchanged conclusion)

Phone “AI” products in 2026 are almost all cloud chat with optional on-device STT/TTS. PocketPal and similar apps run one local GGUF or one ONNX voice. None of the reviewed consumer shells combine concurrent local chat + vision + image diffusion + neural TTS + YAMNet/BirdNET/flora + offline conversation translate + a logged cave-mouth sentry on a Snapdragon 888 without a vendor account. That is the Mk I claim. It is an instrument, not a network appliance.

14. Revision history

Rev	Date	Notes
0.5.4	4 Sep 2026	Translator, horizon, NOAA/OpenSky, FAA warning
0.6.4	5 Sep 2026	G-meter, GPS speed, Sat/Stars, civic watch, first-run fetch
0.6.8	6 Sep 2026	SOL SYSTEM three-step cycle, SWPC + HamQSL + NWS, scrollable earth pane
0.6.14	8 Sep 2026	Sentry weapon watch, continuous III, IPL + Kokoro kill, dwell sliders, audio soundscape watch

End of Rev 0.6.14. Keep this file with the APK and the project zip. The on-device Help link opens a short briefing; this document is the full operations text.